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DAPM-CDR: A domain adaptation prompting model for drug response prediction

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ABSTRACT

The rising incidence and mortality rates of cancer present significant challenges to global health. Variations in tumor growth rates and treatment responses have revealed the limitations of traditional therapies, highlighting the urgent need for predictive models for cancer drug responses based on computational methods. Current drug response prediction methods often fall short in accurately predicting responses for rare cancers with limited data. To address these issues, we propose a domain adaptation prompting model for drug response prediction, DAPM-CDR. DAPM-CDR is trained with cancer types with rich response data of cancer cell lines, integrating both common and specific information into prompts through contrastive learning, to transfer knowledge to target tasks that with sparse data on cancer cell line responses, thereby enhancing the generalization capabilities across various cancer types. The experiment results demonstrate that DAPM-CDR outperforms several competitive methods in predicting drug responses of cell lines, particularly excelling with data from rare cancers and showing significant performance enhancements.

1. Introduction

The rising incidence and mortality rates of cancer pose a significant challenge to global health, with the disease responsible for approximately one-sixth of all deaths worldwide [1]. One of the primary obstacles in treating cancer is its heterogeneity; cancers vary widely in their growth rates, modes of spread, and responses to drugs. The variability significantly undermines the effectiveness of standard treatment approaches [2], emphasizing the urgent need for personalized drug selection [3].

Cell lines, particularly cancer cell lines, provide scientists with a controlled model to study the intricate relationship between disease mechanisms and drug actions. By examining how drugs interact with cancer cell lines under controlled experimental conditions, researchers can explore the complex characteristics of cancer and its potential development of drug resistance. These studies are usually guided by the unique genetic and molecular features of cancer cells, and they help to elucidate which gene mutations, changes in expression patterns, and other intracellular processes may affect treatment responses. The ultimate research goal is to predict different patients' reactions to treatment plans and customize treatment plans for each patient, conforming more closely to each patient's unique characteristics of cancer.

Predicting drug responses has been crucial in personalized medicine, with methodologies evolving significantly over time. Initially, studies focused on traditional machine learning methods. For instance, Rahman et al. [4] improved random forest models by adjusting weights to account for sample heterogeneity, addressing it as an underlying classification challenge. Meanwhile, Ammad-Ud-Din et al. [5] employed Bayesian linear regression, selecting features based on their biological relevance. Their focus on kinase panel data aimed to enhance prediction accuracy. Additionally, Wang et al. [6] developed a matrix factorization method grounded in similarity, integrating drug chemical structures and gene expression profiles to predict cell line responses more accurately.

The field of drug response prediction has witnessed considerable advancement in recent years, thanks in part to the emergence of extensive datasets like the Cancer Cell Line Encyclopedia (CCLE) [7] and the Genomics of Drug Sensitivity in Cancer (GDSC) [8]. These developments, combined with the expansive potential of deep learning technologies, have transformed cancer research and personalized medicine. Choi et al. [9] introduced a novel deep neural network model called RefDNN, which leverages the principle that similar chemical substances typically have similar biological effects. It utilizes a series of samples known as reference drugs to predict drug response labels

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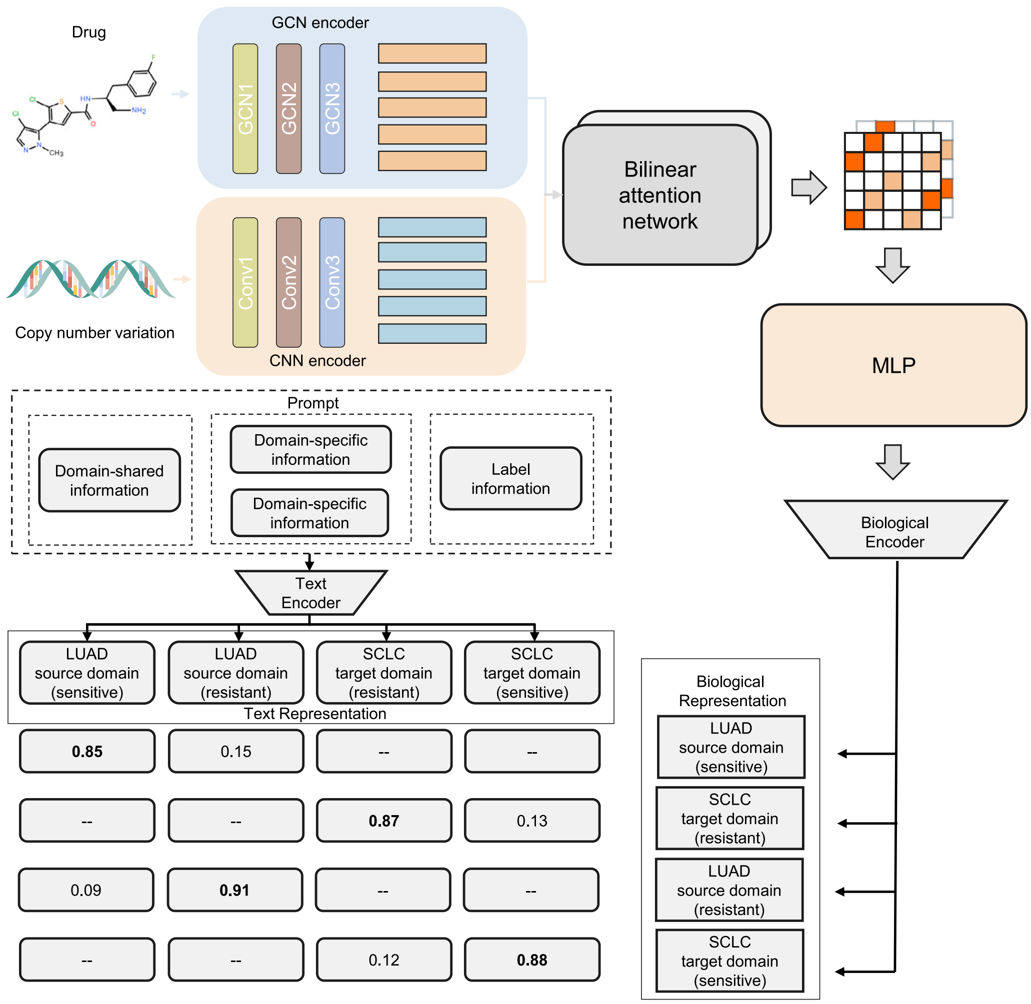

Fig. 1. The overall architecture of the proposed DAPM-CDR model.

by learning representations of high-dimensional gene expression vectors and drug molecule structure vectors based on reference drugs. Oloulade et al. [10] designed AutoCDRP, aimed at automatically predicting cancer drug response using graph neural networks. It employs surrogate modeling techniques to efficiently determine the optimal GNN architecture. By predicting the performance of different GNN architectures, AutoCDRP can select the most suitable architecture based on performance evaluation. Li et al. [11] introduced the MMCL-CDR model for multimodal cancer drug response prediction. The model incorporates various data types including copy number variation, gene expression, cell morphology images, and drug chemical structures to enhance prediction accuracy. It pioneers a novel approach to integrated multi-omics and multimodal modeling of drugs and cell lines. Zhao et al. [12] proposed the MSDRP model aiming to predict drug response, considering the relationships between drugs and biological entities such as targets, diseases, and side effects. The MSDRP model employs an interaction module to capture the interactions between drugs and cell lines, integrating various associations and interactions between drugs and biological entities through a similarity network fusion algorithm. Experimental results demonstrate the outstanding performance of the model in predicting new drug responses.

Despite the immense potential demonstrated by deep learning methods in predicting cancer drug responses, there are still significant challenges in practical implementation. First and foremost, many deep learning models function as "black box" models, and the complexity and lack of transparency in their decision-making processes often impact the trust and acceptance that healthcare providers and patients have towards predicted outcomes. Secondly, the application of these models in predicting drug responses for different types of cancer is facing substantial difficulties. Collecting high-quality cell line data for certain specific types of cancer might be challenging, possibly due to the difficulty in cultivating certain types of cell lines, or because of unequal resource allocation or less attention paid to certain cancers resulting in the insufficiency of the quality and quantity of collected data. Moreover, realistic clinical conditions might introduce new information about drugs and the genome that is not part of the model's training set. It means that the model has to have capability to generalize across domains in order to handle new types of data, and to provide reliable predictions for various types of cancer.

To address existing challenges in predicting cancer drug responses, we introduce a Domain Adaptation Prompting Model for Cancer Drug Response Prediction (DAPM-CDR). DAPM-CDR leverages a domain adaptation strategy, organizing data based on cancer types. Specifically, it treats rich data cancer types as the "source domain" for initial learning and rare data types as the "target domain" for subsequent applications. DAPM-CDR is designed to understand both shared and unique features across different cancers by using specialized training prompts that incorporate both general and cancer-specific information. When combined with the principles of contrastive learning, these prompts empower DAPM-CDR with the nuanced ability to recognize patterns in drug responses that transcend individual cancer types, as well as those that are cancer-specific. Further distinguishing DAPM-CDR is its incorporation of a bilinear attention mechanism, which is particularly adept at parsing copy number variation (CNV) data,

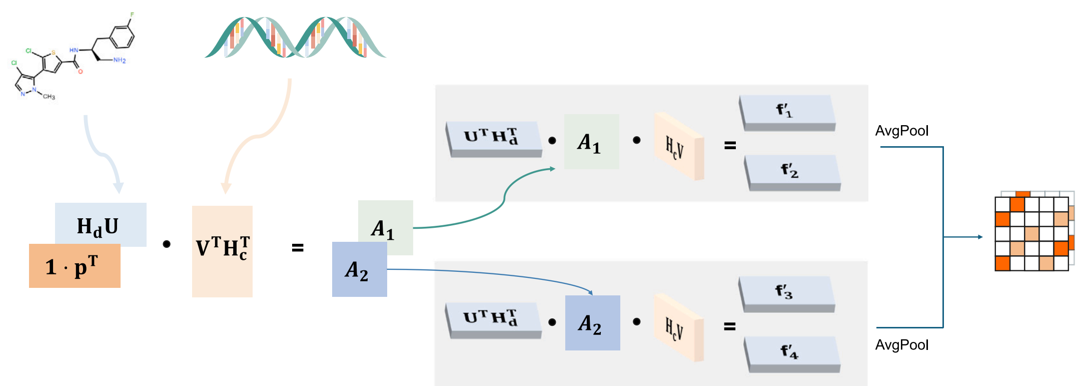

Fig. 2. Details of the bilinear attention network. $\mathbf{H}_d$ and $\mathbf{H}_c$ represent the encoded drug information and copy number variation of cell line information, respectively, obtained through GCN and CNN. The data are transformed via the learnable matrices $\mathbf{U}$, $\mathbf{V}$, and learnable weight vector $\mathbf{p}$, generating the attention mapping matrices $\mathcal{A}_1$ and $\mathcal{A}_2$. These two mapping matrices reveal the strength of interaction between drug substructures and copy number variation of cell line. Following linear transformations with matrices $\mathbf{U}$ and $\mathbf{V}$, along with subsequent average pooling layer processing, we obtain the final drug–cancer cell matching representation.

spotlighting critical mutated genes that play pivotal roles in cancer development and progression. Importantly, it draws connections between these genes and specific enzymes that are targeted by cancer drugs, facilitating a nuanced matching process between drugs and genes. The capability not only enhances the model's predictive accuracy but also provides interpretable insights into the underlying rationale for its predictions.

Experimental evaluations conducted on publicly available datasets show that the DAPM-CDR model demonstrates superior performance compared to baseline models, particularly in predicting drug responses for rare types of cancer where data scarcity is a common issue. Furthermore, through visual analysis comparing the data distribution before and after using our model, we observed a significant homogenization of data distribution across different types of cancer. DAMP-CDR standardizes data distributions across different cancer types to facilitate more consistent representation learning and accurate drug response prediction, demonstrating its generalization capabilities.

2. Materials and methods

The cancer drug response prediction model, DAPM-CDR, is depicted in Fig. 1. It consists of five components: drug representation learning, cell line representation learning, cell line drug response learning, cross-domain prompt learning, and domain adaptation contrastive learning. Data are divided into source and target domains based on the type of cancer. During the training phase, data from both the source domain and target domain are used as inputs to the model, where the source domain data is equipped with real labels and the target domain data utilizes a pseudo-labeling strategy. Specifically, a GCN is employed initially to handle the structural analysis of drug molecules, and a CNN is used to process the Copy Number Variation (CNV) data of cell lines. A bilinear attention mechanism then integrates these two processed datasets to form a unified representation of the interaction between drugs and cancer cell lines (Shown in Fig. 2). Subsequently, custom-designed prompts are fed into a text encoder to capture the representation of the prompts. These drug-cell line matching representations are then paired with the corresponding prompt representations to form positive pairs, while all other representations form negative pairs. Leveraging a contrastive learning mechanism, the model is trained to minimize the distance between positive pairs and maximize the distance between negative pairs to bolster predictive performance, as illustrated in Fig. 3. Notably, for the target domain data, the pseudo-labeling framework assists in guiding the contrastive learning process.

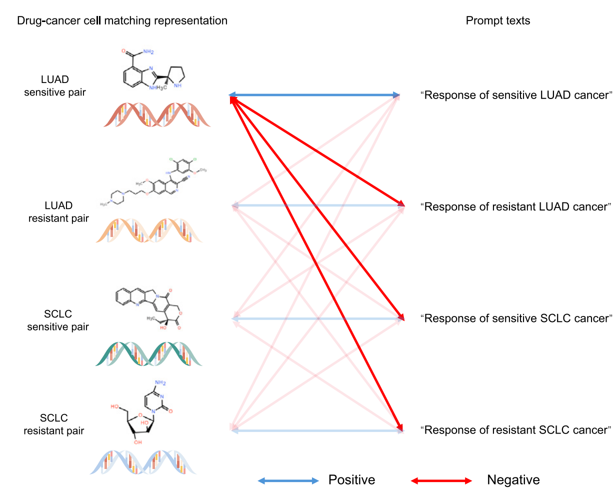

Fig. 3. Domain adaptation via prompt contrastive learning. The left side of the diagram displays the drug–cancer cell matching representation, which shows chemical information. On the right side, manually crafted prompt texts are presented. The aim is to minimize the distance between the drug–cancer cell matching representation and its corresponding prompt texts (depicted in blue) while increasing the mismatch distance (illustrated in red). For additional details, see Section 2.6.

2.1. Problem definition

Initially, we categorize the data into two parts based on cancer type: source domain with abundant training data $\mathcal{D}_s = \{(\mathbf{x}_i^s, y_i^s)\}_{i=1}^{N_s}$ and target domain with unlabeled data $\mathcal{D}_u = \{(\mathbf{x}_i^u)\}_{i=1}^{N_u}$, where N_s and N_u signify the sizes of the source and target domain datasets, respectively. The variables $\mathbf{x}_i^s$ and $\mathbf{x}_i^u$ represent the cancer drug response data from the source and target domains, where each $\mathbf{x}$ consists of a drug and cell line pair and y_i^s represents the labels from the source domain.

2.2. Drug representation learning

For representing drugs, we utilize SMILES strings sourced from PubChem. Previous studies have directly input SMILES into the model, potentially compromising accuracy by disregarding the inherent structural information embedded in its one-dimensional format. To address the issue, SMILES is constructed into a graph data structure $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = (v_1, v_2, \ldots, v_a)$ and $\mathcal{E} = (e_1, e_2, \ldots, e_b)$, where a is the number of atoms in a drug, and b is the number of chemical bonds, enhancing drug information representation. Each graph node v corresponds to an atom. It is encoded into a 74-dimensional vector using DGL-LifeSci package functions [13], which includes attributes such as atom degree, type, formal charge, implicit hydrogen (H) count, atom hybridization, radical electron count, total H count, and whether the atom is aromatic. To accommodate molecules of different sizes, we establish a maximum node count, Θ_n, and apply zero-padding to molecules with less than Θ_n nodes. An additional dimension is introduced to differentiate between original and virtual nodes; it is set to 1 for actual nodes and 0 for the virtual ones. As a result, each drug molecule is represented by a matrix $\mathbf{X}_d \in \mathbb{R}^{\Theta_n \times 75}$. We begin by applying linear transformations to each drug to obtain dense matrices:

$$\mathbf{M}_d = \mathbf{X}_d \mathbf{W}_0, \tag{1}$$

where $\mathbf{M}_d \in \mathbb{R}^{\Theta_n \times D_a}$, $\mathbf{W}_0$ is the learnable matrix for the linear transformation. Subsequently, $\mathbf{M}_d$ is fed into a GCN model, which effectively captures drug structural information while preserving node-level details:

$$\mathbf{H}_d^{l+1} = \sigma(\mathrm{GCN}(\tilde{\mathbf{A}}, \mathbf{W}_d^l, \mathbf{b}_d^l, \mathbf{H}_d^l)), \tag{2}$$

where σ is the non-linear activation function, $\mathbf{W}_d^l$ is a learnable weight matrix of lth layer, $\mathbf{b}_d^l$ is a bias vector, $\tilde{\mathbf{A}} = \mathbf{A}_d + \mathbf{I}$ is the adjacency matrix with self-connections, and $\mathbf{H}_d^l$ ($\mathbf{H}_d^0 = \mathbf{M}_d$) is the representation of nodes (atoms) in lth layer.

2.3. Cell line representation learning

Copy number variations (CNVs) are used as the initial representation of cancer cell lines. CNVs, alterations in the gene copy number, significantly affect cancer development and progression by influencing gene expression levels. The representation is formalized as a matrix $\mathbf{X}_c \in \mathbb{R}^{c \times D_c}$, where c represents the number of cancer cell lines under study, and D_c is the number of genes considered. The matrix captures the copy number variations across different cell lines, providing a genomic landscape of the cancer cells. The matrix $\mathbf{X}_c$ undergoes transformation via the embedding matrix $E \in \mathbb{R}^{\Theta_c \times D_e}$, where Θ_c represents the maximal diversity among CNVs, indicating the broadest range of CNV presence across genes. The parameter D_e represents the dimensions of the potential embedding space, effectively transforming the CNV data into a lower-dimensional, dense representation. Each cancer cell line, after embedding with E, producing a new matrix representation $\mathbf{M}_c \in \mathbb{R}^{D_c \times D_e}$. The transformation aims to encode the CNV information into a more informative and computationally manageable format. The transformed matrix $\mathbf{M}_c$ is subsequently processed by a CNN module consisting of three consecutive one-dimensional (1D) convolutional layers:

$$\mathbf{H}_c^{l+1} = \sigma(\mathbf{CNN}(\mathbf{W}_c^l, \mathbf{b}_c^l, \mathbf{H}_c^l)), \tag{3}$$

where $\mathbf{H}_c^l$ ($\mathbf{H}_c^0 = \mathbf{M}_c$) represents the representation of cancer cell lines in lth layer, $\mathbf{W}_c^l$ is a learnable weight matrix of lth layer, and $\mathbf{b}_c^l$ is the bias vector.

2.4. Cell line drug response learning

At this stage, a bilinear attention network is employed to examine interactions between drugs and cell lines. After the previous steps, we obtain the drug representation $\mathbf{H}_d = \{\mathbf{h}_d^1, \mathbf{h}_d^2, \ldots, \mathbf{h}_d^N\}$ and cell line representation $\mathbf{H}_c = \{\mathbf{h}_c^1, \mathbf{h}_c^2, \ldots, \mathbf{h}_c^M\}$, where N represents the number of atoms in the drug, and M represents the number of genes of cell line. We define an attention mapping matrix $\mathcal{A} \in \mathbb{R}^{N \times M}$ as follows:

$$\mathcal{A} = ((\mathbf{1} \cdot \mathbf{p}^\top) \odot \sigma(\mathbf{H}_d \mathbf{U})) \cdot \sigma(\mathbf{V}^\top \mathbf{H}_c^\top), \tag{4}$$

where $\mathbf{p} \in \mathbb{R}^K$ is a learnable weight vector, $\mathbf{U} \in \mathbb{R}^{D_a \times K}$ and $\mathbf{V} \in \mathbb{R}^{D_e \times K}$ are learnable weight matrices, $\mathbf{1} \in \mathbb{R}^N$ signifies a ones vector, and $\odot$ denotes the Hadamard product (element-wise multiplication). To better represent the bilinear joint final representation $\mathbf{f}$, we define a pre-joint representation $\mathbf{f}' \in \mathbb{R}^K$ as follows:

$$\mathbf{f}'_k = \sigma(\mathbf{H}_d \mathbf{U})_k^\top \cdot \mathcal{A} \cdot \sigma(\mathbf{H}_c \mathbf{V})_k, \tag{5}$$

where k on the left side of the equation represents the kth intermediate representation, and k on the right side represents the column index of the matrix. During this stage, we use the parameters as in to reduce computational complexity and mitigate overfitting. Subsequently, we introduce an average pooling layer to obtain the final joint representation:

$$\mathbf{f} = \mathrm{AvgPool}(\mathbf{f}', s), \tag{6}$$

where s represents the pooling window size, which is also the stride. Next, we compute multiple joint representations and sum them to obtain the final multi-head joint representation. Subsequently, the multi-head joint representation is fed into a MLP to produce the ultimate output of the module dedicated to learning interactions between drug–cancer cell pairs. The final drug–cancer cell matching representation $\mathbf{X}$ can be calculated as follows:

$$\mathbf{X} = \mathrm{MLP}(\sum_{i=1}^{T} \mathbf{f_i}), \tag{7}$$

where T represents the number of attention heads.

Additionally, the attention mapping matrix represents the associations between drug and copy number variation of cell line substructures, reflecting potential causes of cancer drug response. To facilitate interpretable analysis of the model, we define the interpretability matrix I as follows:

$$\mathrm{I} = \sum_{i=1}^{T} \mathcal{A}_i. \tag{8}$$

2.5. Cross-domain prompt learning

While we have successfully constructed a matching representation model between drugs and cancer cells, challenges remain in accurately predicting cancer drug responses in practical situations. Especially in predicting drug responses for different types of cancer, it is critical to address the issue of domain disparities to enhance the accuracy of our predictions. Therefore, while learning the interactions between cell lines and drugs, our research also focuses on overcoming the difficulties in domain adaptation. Currently, most unsupervised domain adaptation (UDA) methods focus on learning invariant features by minimizing differences between domains. However, the approach might overlook crucial information due to the unique features present in each domain. Considering the significant success of prompts in the natural language processing and computer vision fields [14,15], we propose a novel learnable prompt design approach. It approach emphasizes both shared and distinct features across domains. In selecting the prompt design, given the fixed and limited number of domains in our experiment, we choose a manual prompt design method, which enables precise description of different domains. Specifically, these manually designed prompt texts adhere to a unified format, defined as follows:

$$\mathbf{t}_z^d = [\mathbf{v}]_1[\mathbf{v}]_2 \cdots [\mathbf{v}]_{M_1}[\mathbf{d}]_1^d[\mathbf{d}]_2^d \cdots [\mathbf{d}]_{M_2}^d[\mathbf{CLASS}]_z, \tag{9}$$

where $[\mathbf{v}]$ is domain-shared information, which is shared across all domains, $[\mathbf{d}]$ is domain-specific information, shared only within specific domains, $[\mathbf{CLASS}]$ is the label information, indicating whether the cancer drug response is sensitive or resistant. $d \in \{s, u\}$ represents different domains, where s denotes the source domain and u represents the target domain. Similarly, $z \in Z$ indexes response labels, where $Z = \{sensitive, resistant\}$. M_1 and M_2 specify the token counts for domain-shared information and domain-specific information, respectively.

2.6. Domain adaptation contrastive learning

To enhance the usefulness of manually designed prompt texts and the final drug–cancer cell matching representation, a contrastive learning approach is adopted for domain adaptation. Given that the prompts are in textual form and not directly computable, they are converted into computable vectors via text encoding. Specifically, we define a text encoder, denoted as $f(\cdot)$, based on a Transformer model. The manually designed prompt texts are input into $f(\cdot)$ to generate fixed vectors in the word embedding space. Subsequently, these vectors are used for the contrastive training of drug–cancer cell pairs with corresponding text matches. For example, in the domain of lung adenocarcinoma (LUAD), the final drug–cancer cell matching representation $\mathbf{X}_i$ is a sensitive pairing of drug and cancer cell line, where the input text $\mathbf{t}_i$ is "a response of a sensitive LUAD cancer". We consider $\mathbf{X}_i$ and $\mathbf{t}_i$ as matching pairs, while $\mathbf{X}_i$ and $\mathbf{t}_j$ represent mismatched pairs. Cosine similarity is utilized to minimize the distance between $\mathbf{X}_i$ and $\mathbf{t}_i$, and to maximize the distance between $\mathbf{X}_i$ and $\mathbf{t}_j (i \neq j)$. The alignment brings the prompt and drug–cancer cell matching representation closer together spatially.

Following transforming target domain $\mathcal{D}_s = \{(\mathbf{x}_i^s, y_i^s)\}_{i=1}^{N_s}$ and unlabeled data $\mathcal{D}_u = \{(\mathbf{x}_i^u)\}_{i=1}^{N_u}$ through the aforementioned steps, we obtain a set of training samples $\{(\mathbf{X}_i^s, y_i^s)\}_{i=1}^{N_s}$ and unlabeled samples $\{(\mathbf{X}_i^u)\}_{i=1}^{N_u}$. Consequently, the probability of belonging to class z can be calculated as follows:

$$P\left(\hat{y}_i^s = z | \mathbf{X}_i^s, \mathbf{t}_z^s\right) = \frac{\exp(\left\langle f\left(\mathbf{t}_z^s\right), \mathbf{X}_\mathbf{i}^s\right\rangle / t)}{\sum_{d \in \{s,u\}} \sum_{j=1}^{Z} \exp\left(\left\langle f\left(\mathbf{t}_j^d\right), \mathbf{X}_\mathbf{i}^s\right\rangle / t\right)}, \tag{10}$$

$$P\left(\hat{y}_i^u = z | \mathbf{X}_i^u, \mathbf{t}_z^u\right) = \frac{\exp\left(\left\langle f\left(\mathbf{t}_z^u\right), \mathbf{X}_\mathbf{i}^u\right\rangle / t\right)}{\sum_{d \in \{s,u\}} \sum_{j=1}^{Z} \exp\left(\left\langle f\left(\mathbf{t}_j^d\right), \mathbf{X}_\mathbf{i}^u\right\rangle / t\right)}, \tag{11}$$

where t is the temperature parameter, and $\langle \cdot, \cdot \rangle$ denotes the cosine similarity. Based on the above equations, the loss function for the source domain is introduced as follows:

$$\mathcal{L}_s = -\frac{1}{N_s} \sum_{i=1}^{N_s} \log P\left(\hat{y}_i^s = y_i^s\right), \tag{12}$$

where y_i^s represents the ground-truth of the source domain.

In UDA tasks, the lack of labels in the target domain makes it impossible to train the model using labels. To address this issue, a method involving pseudo-labels is introduced for training. The class with the highest predicted probability is selected as the pseudo-label y^u for the training data $\mathbf{X}^u$, defined as:

$$y^u = \arg\max_z P(\hat{y}^u = z | \mathbf{X}^u). \tag{13}$$

Based on our methods approach, pseudo-label information for the target domain is obtained, and the loss function is constructed using pseudo-labels as:

$$\mathcal{L}_u = -\frac{1}{N_u} \sum_{i=1}^{N_u} \mathbb{I}\{P(\hat{y}_i^u = y_i^u | \mathbf{X}_i^u) \geq \tau\} \log P\left(\hat{y}_i^u = y_i^u | \mathbf{X}_i^u\right), \tag{14}$$

where $\mathbb{I}\{\cdot\}$ is the indicator function and τ is the pseudo-label threshold. In other words, when the predicted probability not exceeds τ, it is not calculated into the loss function. The total loss function $\mathcal{L}$ can be calculated as follows:

$$\mathcal{L} = \mathcal{L}_s(\mathcal{D}^s) + \kappa \mathcal{L}_u(\mathcal{D}^u), \tag{15}$$

where κ represents the coefficient balancing the source and target domain contributions.

3. Experiments

In this section, we evaluate the performance of our DAPM-CDR model in predicting cancer drug responses using a benchmark dataset. Our analysis focuses on six key questions: (i) Does the DAPM-CDR model outperform existing baseline models? (ii) How effective are the prompt module and its design in enhancing cancer drug response (CDR) predictions? (iii) Does the pseudo-labeling approach improve performance in domain transfer tasks? (iv) What impact does the bilinear attention network have on the model's overall performance? (v) How interpretable are the results generated by our bilinear attention network? (vi) How effective is our model in leveraging common cancer knowledge to enhance the prediction of rare cancer types?

3.1. Datasets

In this section, we assess the DAPM-CDR model's performance using datasets compiled from two publicly available databases: the Genomics of Drug Sensitivity in Cancer (GDSC) and PubChem. Details of these datasets are presented in Table 1. **GDSC** serves as an extensive resource for exploring the sensitivity of cancer cells to drugs and their molecular responses, offering data on over 1000 cancer cell lines. It includes IC50 values, targeted pathways, gene expression profiles, and covers diverse cancer types such as lung adenocarcinoma, breast cancer, and colorectal cancer, thereby facilitating cancer biology and therapeutic studies. **PubChem**, with its comprehensive chemical data, including chemical names, molecular formulas, and SMILES strings, is our principal source for drug chemical structure information.

For our primary experiment, we selected Lung Adenocarcinoma (LUAD) as the source domain and Small Cell Lung Cancer (SCLC) as the target domain, compiling 3,200 drug–cancer cell interaction pairs across these domains. Throughout the training phase, the model is fed with all labeled LUAD data alongside the unlabeled SCLC data. The trained model is then deployed to predict the drug response in SCLC.

Additionally, to test the model's domain adaptation capacity for less prevalent cancers, we kept LUAD as the source domain and incorporated two additional rare cancer types as new target domains: Adrenocortical Carcinoma (ACC) and Chronic Lymphocytic Leukemia (CLL). For these, we extracted 217 and 426 drug–cancer cell interaction pairs from ACC and CLL, respectively, while the data from the source domain (LUAD) was kept constant at 3,200 pairs.

3.2. Experiment settings

3.2.1. Baselines

To evaluate the performance of our DAPM-CDR model, we conducted comparisons against a set of baseline models:

- **GraphCDR** [16]: GraphCDR integrates multiple omics datasets into a CNN framework for cancer drug response (CDR) prediction. It uses contrastive learning as a regularization technique to improve its generalization capabilities.
- **NIHGCN** [17]: NIHGCN develops a predictive framework to address disparities between cell lines and drug nodes. It leverages diverse levels of neighborhood interaction (NI) within a heterogeneous graph convolutional network, resulting in improved performance in predicting associations between new drugs and cell lines.
- **GraphDRP** [18]: GraphDRP represents drugs as molecular graphs to capture atomic bonds and describes cell line genetic mutations as binary vectors. It then predicts interactions between drugs and cell lines using a fully connected layer.
- **DeepTTC** [19]: DeepTTC is an end-to-end deep learning model that utilizes transformers to learn drug representations and employs a multi-layer neural network to predict associations between cell lines and drugs.

3.2.2. Evaluation metrics

We employ two principal evaluation metrics to assess the performance of the classification models: **AUC** (Area Under the ROC Curve) and **AUPR** (Area Under the Precision–Recall Curve).

AUC represents the area under the ROC curve, which plots the true positive rate (TPR) against the false positive rate (FPR) at various classification thresholds. The value of **AUC** ranges between 0 and

Table 1
Statistics of datasets.

Data type	Counts	Address
Drug SMILES	415 drugs	https://pubchem.ncbi.nlm.nih.gov/
Copy Number Variation	969 cell lines - 24502 genes	https://www.cancerrxgene.org/

Table 2
The comparison results of DAPM-CDR and baseline methods.

Method	AUC	AUPR
NIHGCN	0.7605	0.7403
GraphCDR	0.8063	0.8003
GraphDRP	0.8967	0.7728
DeepTTC	0.8953	0.8957
our model	**0.9167**	**0.9185**

Table 3
Results of the ablation study.

Method	AUC	AUPR
w/o Prompt	0.8340	0.8341
w/o Pseudo-labels	0.8844	0.9049
w/o bilinear attention network	0.8882	0.9013
Only General Prompt	0.8847	0.8971
Full model	**0.9167**	**0.9185**

1, with a higher value indicating better performance, serving as a comprehensive measure for the model.

AUPR calculates the area under the precision–recall curve, focusing on the predictive performance of positive instances, which aligns closely with the principles of precision medicine. Similar to AUC, a higher AUPR value indicates better performance.

3.2.3. Experimental setup

In the context of cancer drug response prediction, drug interactions with cancer cell lines are often quantified using IC50 values, where lower values indicate a stronger inhibitory effect. It is a common practice in numerous studies to dichotomize the response level into 'sensitive' or 'resistant' based on these IC50 metrics. Following the experimental framework by Brubaker et al. [20], a threshold of 0.1 was established for cancer drug response. Thus, responses with an IC50 value at or below 0.1 are classified as 'sensitive', while those above are considered 'resistant'.

Concerning the DAPM-CDR model's learning rates, disparate values are assigned for the source and target domains, being 0.3 and 1e−4 respectively, and a decay of 50% is implemented every 2000 batches. Hyperparameters t and τ are set at 1 and 0.8, respectively, and the domain coefficient κ is consistently at 0.5. In the drug feature extraction module, a GCN is utilized, structured with 3 layers, each having dimensions of [128, 128, 128]. The cell line feature extraction employs a three-layer CNN with dimensions [128, 128, 128] and kernel sizes [3, 6, 9]. Parameters for these components retain the settings referenced in the methodology of other baseline studies.

3.3. Experiment results

3.3.1. Comparison with baselines

We conducted a comparative analysis of the DAPM-CDR model developed with four baseline models. Detailed comparison results, including AUC and AUPR metrics, are presented in Table 2. Specifically, the DAPM-CDR model showed at least a 2% improvement in all comparison metrics, indicating its significant advantage for this prediction task. It is worth mentioning that models trained on graph-structured data, such as GraphCDR and NIHGCN, performed relatively poorly. The phenomenon is likely related to these models' limited ability to recognize and learn about novel nodes when processing cross-domain data, thereby restricting their effectiveness in cross-domain applications. Given these findings, the DAPM-CDR approach in this study has been validated theoretically and has demonstrated high-efficiency performance in practical applications for complex drug-compound reaction prediction tasks, providing an important reference for the design of similar models in the future.

3.3.2. Ablation study

To evaluate the contribution of each component, we perform ablation studies to confirm their effectiveness. The strategies are as follows: (1) *w/o* Prompt: We exclude the prompt generation module and directly feed the vectors from the bilinear attention network into an MLP for classification. (2) *w/o* Pseudo-labels: We eliminate the pseudo-labeling strategy from our experiments. (3) *w/o* bilinear attention network: We excluded the bilinear attention network strategy from our experiments and concatenated the GCN and CNN-obtained representations of drugs and cell lines, which were then fed into an MLP. (4) Only General Prompt: We implement a uniform prompt for all categories, not differentiating between specific domains.

Based on the experimental results shown in Table 3, several key conclusions can be drawn: (1) The removal of the prompt module ("w/o Prompt") significantly degraded the model's performance, as evidenced by the lower AUC and AUPR values. This indicates that the prompt is a vital component for effective generalization across diverse domains. Its absence likely led to overfitting within the source domain, highlighting the complexity and interplay of the drug and cancer cell encoding modules along with the bilinear attention mechanism. (2) The exclusion of pseudo-labels ("w/o Pseudo-labels") also negatively impacted the model's performance. It underscores the critical role of pseudo-labeling in enhancing the model's generalization abilities. The strategy appears to be a valuable asset in improving the overall performance of the model. (3) The exclusion of the bilinear attention network ("w/o bilinear attention network") led to a significant decline in model performance. The outcome underscores the importance of the bilinear attention network in capturing the interaction between drugs and cell lines. The absence of the bilinear attention network may hinder the model from fully leveraging the potential value in the dataset and weaken its explanatory capability. (4) The use of a generic prompt ("Only General Prompt") did not match the performance of the more tailored approach. The finding suggests that domain-specific prompts are essential in capturing the intricacies of different categories, thereby leading to improved outcomes. (5) The full model, which incorporates all the key components, consistently excelled in comparison to the other variations. It underscores the synergistic effect of combining the prompt module, pseudo-labeling, and domain-specific prompts. The results validate the effectiveness of the overall approach and emphasize the importance of maintaining a holistic design in model development.

3.3.3. Parameter sensitivity analysis

To evaluate the robustness of our model, we conducted a parameter sensitivity analysis. Specifically, we modified the value of the domain coefficient κ and examined how this change affects model performance. For κ, we tested values of 0.1, 0.3, 0.5, 0.7, and 0.9, and the results are shown in Fig. 4. These results indicate that with the increase of κ, the model's AUC and AUPR evaluation indicators first increased and then decreased. When κ is set to 0.5, the model performance is optimal.

In addition, we further studied the impact of pseudo-label coefficient τ on model performance. We adjusted the pseudo-label coefficient

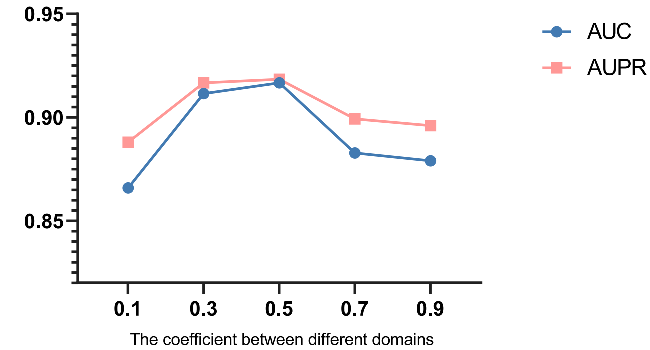

Fig. 4. Results with different hyperparameter κ.

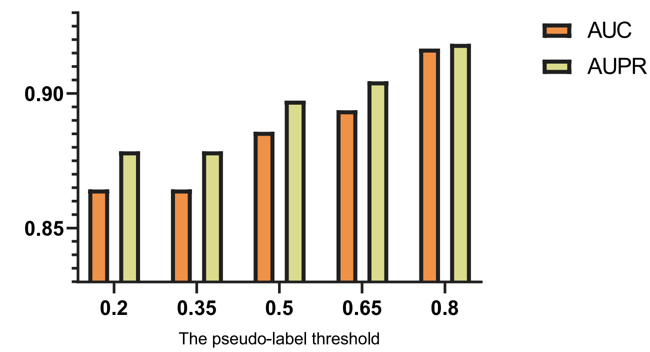

Fig. 5. Results with different hyperparameter τ.

to 0.2, 0.35, 0.5, 0.65, and 0.8, and the results are shown in Fig. 5. As the pseudo-label coefficient increased, the accuracy of the model improved. The performance improvement should be attributed to a higher pseudo-label coefficient raising the threshold of the indicator function, ensuring that only high-confidence data is marked with pseudo-label and participates in the process of model backpropagation. The strategy significantly improved the model's performance. Another noteworthy finding is that when the pseudo-label coefficient is set between 0.2 and 0.35, the model performance shows a similar trend. This could be because below a certain threshold, the pseudo-label coefficient may unintentionally include all prediction results, thereby reducing the model's adaptability across different domains.

3.3.4. Interpretation analysis

Our model operates with a bilinear attention module to facilitate result interpretation. We focus on the attention scores derived from , symbolizing the attention weight given to the correlation between each atom of the drug under consideration and the copy number variation of cell line. As our primary objective lies in unraveling this relationship, we collate the scores for each atom of the drug into a matrix. The matrix then visualizes the interaction between the drug and the copy number variation of cell line in the context of the cancer cell lines. Taking into account the expansive dataset, we cherry-picked a few representative examples and drew up a heatmap, which is showcased in Fig. 6.

For Afuresertib and the NCI-H2342 cancer cell line, the attention score for the AKT2 gene in copy number variation is the highest. AKT2 is known to be highly expressed in various human cancers, particularly in non-small cell lung cancer (NSCLC). Its overexpression is closely associated with promoting proliferation and metastasis of NSCLC cells, leading to poor prognosis in lung adenocarcinoma patients. Additionally, AKT2 influences the growth, migration, and invasion of adenocarcinoma cells in the lung by regulating the cell cycle, promoting epithelial–mesenchymal transition (EMT), and increasing the expression of matrix metalloproteinases (MMPs). Overexpression

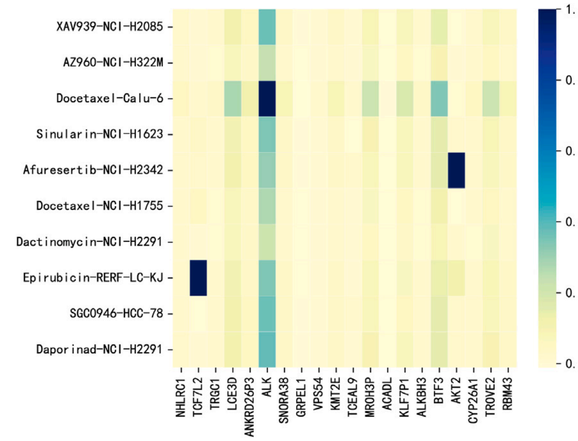

Fig. 6. Visual heatmap of the attention matrix.

of miR-124 can downregulate AKT2 expression, inhibiting lung adenocarcinoma progression. Some studies suggest that the miR-124/AKT2 pathway may become a potential target for new therapies for lung adenocarcinoma [21]. Afuresertib, an AKT inhibitor, has been reported to increase the expression of p21 WAF1/CIP1 and reduce the phosphorylation of Akt substrates, including GSK-3β and FOXO family proteins [22].

In the case of docetaxel and the Calu-6 cell line, the highest attention score was for the ALK gene's copy number variation. ALK consistently acts as an oncogene in lung cancer, particularly NSCLC, accounting for about 4%–6% of lung adenocarcinoma cases. ALK rearrangements lead to abnormal expression and sustained activation of ALK protein kinase activity. The type of lung cancer shows dependency on ALK kinase inhibitors (TKIs), exhibiting high sensitivity to these drugs [23]. Docetaxel is used to treat various cancers, including breast cancer, gastric cancer, and small cell lung cancer.

Additionally, our study also focuses on the TCF7L2 gene, known for its high expression or mutational prevalence across a range of human tumors, including lung cancer [24]. Rice et al. conducted a retrospective analysis of 183 lung cancer patients who underwent commercial NGS sequencing. Additionally, he studies the LUAD and squamous cell carcinoma (LUSC) datasets from The Cancer Genome Atlas (TCGA), uncovering several key findings. Using Transcription Factor Enrichment Analysis (TFEA), they grouped these mutations and analyzed their association with patient survival rates. The results revealed a correlation between mutations related to the transcription factor TCF7L2/TCF4 and lower survival rates in NSCLC patients. Additionally, the study identified a strong link between TCF7L2 pathway activity and poor prognosis in lung adenocarcinoma, suggesting that targeting this pathway could provide new therapeutic avenues to enhance patient outcomes [25]. Moreover, our model identified other key genes implicated in lung cancer pathogenesis, including ALKBH3 [26] and BTF3 [27].

The remarkable interpretability of our model not only underscores its scientific merit but also paves the way for the discovery of novel targets for cancer treatment. This aspect is especially beneficial for the exploration of genes that have yet to be extensively studied, thus revitalizing research in cancer therapeutics. Within our model's scoring system, a number of genes have been identified as having strong links to lung cancer. Notably, although RNF2P1 and RNASEH2CP1 are amongst the highest ranked, comprehensive research into their connection with cancer remains sparse. The gap in knowledge signals an opportunity for future studies to delve deeper and explore new avenues in the fight against cancer.

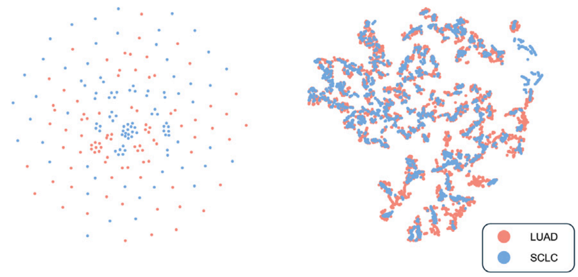

Fig. 7. Visualization illustrates dimension reduction across various cancer cell types. The original dataset distribution is shown on the left, while the right part is learned by our model.

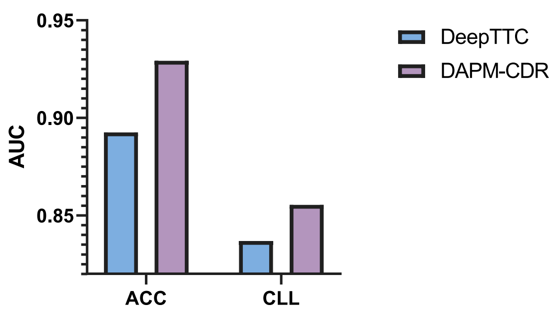

Fig. 8. Domain adaptation in the field of rare cancer.

3.3.5. Visualization analysis

In this study, we utilized the t-SNE algorithm to perform a visual analysis of the performance of our developed DAPM-CDR model in terms of domain transfer. t-SNE is an unsupervised dimensionality reduction technique that is well-suited for visualizing high-dimensional datasets. Its principle lies in preserving the local structural properties of data in the original space to achieve the mapping from high dimensions to lower dimensions. This ensures that sample points with high local structure similarity are closer in the mapped space, while those with greater differences are farther apart.

When processing and visualizing data for two cancer types, LUAD and SCLC, as shown in Fig. 7, it is clear that before transformation, the data points of the two types of cancer are relatively dispersed in the high-dimensional space. After transformation to the low-dimensional space using the t-SNE algorithm, these distributions become more compact and focused. This indicates that our model has good generalization capability, as it can effectively map the data features of different types of cancer to a similar distribution during the transfer process, thereby affirming the effectiveness and accuracy of the DAPM-CDR model in handling data from different domains.

3.3.6. Performance analysis in rare cancers

To evaluate the ability of our model to transfer knowledge from abundant to rare cancer datasets, we focused on two types of cancer: ACC (Adrenocortical Carcinoma) and CLL (Chronic Lymphocytic Leukemia). ACC is a rare type of cancer, whereas CLL has limited available data. Our model demonstrates competitiveness with the baseline model DeepTTC from prior experiments, especially when dealing with rare or data-scarce cancers, as shown in Fig. 8. This result contributes to overcoming the challenges associated with predicting responses in rare cancers within the field of precision medicine.

4. Conclusion

In this research, we present DAPM-CDR, a novel domain adaptation prompting model specifically crafted for enhancing drug response predictions. DAPM-CDR incorporates advanced prompting techniques from Natural Language Processing (NLP), significantly improving its domain transferability and enabling effective generalization across diverse cancer types. By integrating bilinear attention modules, DAPM-CDR addresses the critical challenge of interpretability that is frequently encountered in deep learning approaches, providing a clearer understanding of model decisions.

Through rigorous experimental evaluation, DAPM-CDR distinguishes itself as a leading contender among current cancer drug prediction models, demonstrating its robustness and competitiveness. One of the model's most noteworthy attributes is its ability to be seamlessly transferred to different types of cancer cell line datasets, a capability achieved by training on well-annotated cancer datasets. This adaptability underscores DAPM-CDR's practicality in a broad spectrum of oncological research contexts. Moreover, DAPM-CDR enhances interpretability within the deep learning landscape, a crucial advancement for the scientific community's understanding of model-driven insights. It goes further to identify previously overlooked genes, offering a fresh perspective on potential biomarkers and therapeutic targets. These discoveries pave the way for novel research methodologies that could profoundly influence future cancer treatment paradigms. In essence, DAPM-CDR not only excels in drug response prediction but also acts as a catalyst for pioneering research directions.

Although the DAPM-CDR model has made certain advancements at the current stage, its applicability is somewhat limited. Presently, the model is only capable of transferring data from one type of cancer to another, a constraint that hinders its ability to fully exploit the wealth of latent information in the data. Therefore, future research must prioritize the exploration of effective domain adaptation across various cancer types, which will serve as a critical area of focus. Moreover, existing studies on the DAPM-CDR model are primarily based on cell line data, which curtails its potential for application to a broader spectrum of clinical data. In the future, extending the application of this model to clinical data can not only enhance its practical value substantially but also facilitate its widespread use in actual clinical settings. In summary, future studies should concentrate on addressing the issues of domain adaptation across multiple cancer types and endeavor to apply the DAPM-CDR model to clinical data, with the aim of further improving the model's practicality and efficacy.

CRediT authorship contribution statement

Youhan Sun: Methodology, Investigation. **Guanyu Qiao:** Methodology, Data curation. **Bo Gao:** Validation, Investigation, Formal analysis. **Yang Li:** Writing – original draft, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data and codes are available at https://github.com/catly/DAPM-CDR.

Acknowledgments

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